A Low-Cost Robotic Arm Manipulator Model for Enhancing Student Learning in an Industrial Robots Course

Mohammad Mashagbeh*, Mustafa Al-Khawaldeh†, Sanad Hussein*, Husam AlAbdellat* and Andraws Swidan‡
*The Department of Mechatronics Engineering, The University of Jordan, Amman, Jordan
Email: m.mashagbeh@ju.edu.jo
†The Department of Mechatronics Engineering, Philadelphia University, Amman, Jordan
‡The Department of Computer Engineering, The University of Jordan, Amman, Jordan

2023 IEEE Frontiers in Education Conference (FIE) | 979-8-3503-3642-9/23/$31.00 ©2023 IEEE | DOI: 10.1109/FIE58773.2023.10343313

***Abstract*—Recent years have seen a rise in the importance of the field of robotics, therefore students need to understand its basic concepts. Forward and inverse kinematics, which are essential components of robotic arm control, must be understood in order to design and program a robotic arm manipulator. To help students understand these concepts, this study proposes a low-cost robotic arm model. The robotic arm model is simple to build, inexpensive, and flexible, allowing students to experiment with various configurations and test various kinematic concepts. The model consists of a 3 DOF RRR (3 revolute joints) robot, providing students with a 3D view of the D-H (Denavit–Hartenberg) parameters. As the robot moves, students can observe and visualize parameters such as the twist of the link α and the angle of the joint θ. This improves students' understanding of the concept of D-H parameters. The proposed robotic arm model can be used to teach students about the basics of robotic arm manipulators while also serving as a less expensive substitute for more expensive commercial models. The D-H parameters, forward and inverse kinematic analysis can be explained using the model, which can also help students comprehend these fundamental mathematical concepts. In general, this inexpensive robotic arm model gives students a practical and convenient approach to advance their robotic knowledge and abilities. According to the results of our study, the use of the proposed model in an industrial robots course can greatly enhance students' learning outcomes. The model gives students practical experience and enables them to apply theoretical ideas in a reality, which improves their comprehension of the subject. Also, it was discovered that the model was a useful tool for involving students in the learning process and boosting their motivation to study. Our research implies that, especially in situations where access to expensive equipment is restricted, using a low-cost robotic arm manipulator model can be a useful addition to industrial robots courses.**

***Index Terms*—RRR Robot, D-H parameters, Forward Kinematics, Inverse Kinematics**

I. Introduction

Nowadays, robotics is considered an essential element in our daily life applications. Thus, integrating it in the educational system provides a strong motivation and results in a great improvement in learning speed [1]. Despite its importance, most curricula have not paid attention to teaching robotics concepts for problem solving.

Research has shown that robotics can help people develop problem-solving skills and cope with the challenges and possibilities of its many devices and processes. Robotic direct manipulation environments activate people'to develop more critical thinking skills and conceptual development [2]. Such environments help students learn robotics best through designing, constructing, programming, testing, and optimizing. Conceptual teaching orients students' attention to significant phenomena and concepts, but it is more difficult than delivering information through direct instruction [2].

According to researchers, the use of models and experiments is crucial in education [3]. It stimulates reflection and thinking, which in turn builds well-grounded conceptual knowledge, as opposed to traditional methods for teaching, which are based on lectures, PowerPoint presentations, and readings. Certain robotics concepts are very difficult to understand and interpret, which ultimately negatively influences the students' performance and learning process of students '. Despite the importance of interaction with such teaching environments, little research has been conducted on the significance of integrating robotics models and systems in teaching robotics essential concepts at the university level.

An example of such concepts is forward and inverse kinematics. Forward and inverse kinematics are essential components of robotic arm control and must be understood in order to design and program a robotic arm manipulator. However, students in some cases find solving the forward and inverse kinematics exercises difficult when attempting to realize the geometric operations and movements that pertain to the joints of a serial manipulator [4]. This shows that traditional teaching methods can't contribute significantly to robotics conceptual development and effective engineering skill acquisition because of the lack of hands-on experience [4]. Though they might understand robotics theoretical concepts, they often can't apply them to solving real-life problems. Integrating educational tools such as computer simulations, virtual reality (VR), and virtual laboratories improves students' learning process [5], [6]. Highfeld [7] showed that robotic toys are significant pedagogical tools for problem-solving sessions and developing students' mathematical skills. The use of such toys promotes learning by engaging students with challenging mathematical concepts.

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on October 29,2026 at 19:31:16 UTC from IEEE Xplore. Restrictions apply.

Certain computer applications and models have been designed for teaching forward and inverse kinematics in Robotics [8]–[11]. They enable students to create and manipulate interactive representations of the robotic models, thus they improve their understanding of the required transformations and operations needed to resolve both forward and inverse kinematics exercises [8], [12], [13]. An educational simulation tool for example has been developed in [14] for teaching the kinematics of serial robotic arms. This tool facilitates comparing different robot arm configurations based on various physical parameters and mechanical features. A software has been developed in [14] for performing kinematic and dynamic analyses of robots as well as teaching essential robotics concepts. Furthermore, a low-cost serial manipulator has been proposed in [15] as an excellent platform for teaching robotics concepts. A virtual universal robot has been designed in [16] as a good educational tool for programming and evaluating the inverse kinematics control of the robot and its joints. RoboAnalyzer which has been applied in [17] as a teaching tool of the robotics course is found to greatly improve learning efficiency.

The review shows that despite the great research efforts to integrate technology in teaching robotics and the resultant favorable results, the studies do not show firm evidence that the students' knowledge is sufficient for transferring "virtual simulations" to the "real world" [18], [19]. Hence, this study aims to address the existing research gap by evaluating students' comprehension of D-H parameters in robotics through both conventional teaching methods and the prototype approach.

II. Forward and inverse kinematics

The forward kinematics problem involves establishing the correlation between the individual joints of a robot manipulator and the position as well as the orientation of its tool or end-effector. More formally, given the joint variable values of the robot, the forward kinematics problem aims to calculate the precise position and orientation of the end-effector. In the case of revolute or rotational joints, the joint variables refer to the angles formed between the links. It is important to note that the forward kinematics problem is distinct from the inverse kinematics problem. While forward kinematics deals with determining the end-effector position and orientation based on the joint variables, inverse kinematics focuses on finding suitable joint variable values to achieve a desired position and orientation for the robot's end-effector.

The D-H parameters table is a crucial tool for characterizing the kinematic properties of a robot and establishing the relationship between its joints and end-effector. It involves a systematic approach, including steps such as locating and labeling joint axes, establishing the base frame, determining origins and coordinate axes for each joint, and forming the end-effector frame. Additionally, a full table containing link lengths, link offsets, link twists, and joint angles is created to define the D-H parameters. Using the D-H parameters table, the robot's geometry and kinematics can be found. As a result, this can be used to analyze, manage, and plan robot's movements and operations.

D-H parameters are associated with their coordinate transformations, but understanding their effect on the physical configuration of the robot is difficult. To overcome this hurdle, a 3D visualization model with attached frames of the actual robot would be a better approach. As a case study, we selected the RRR robot to apply our methodology for understanding the D-H parameters. The RRR robot consists of three revolute joints, making it an ideal candidate to investigate the kinematic relationships and D-H parameter determination.

Figure 1 shows a sketch of the RRR robot, while the D-H parameters are shown in Table I.

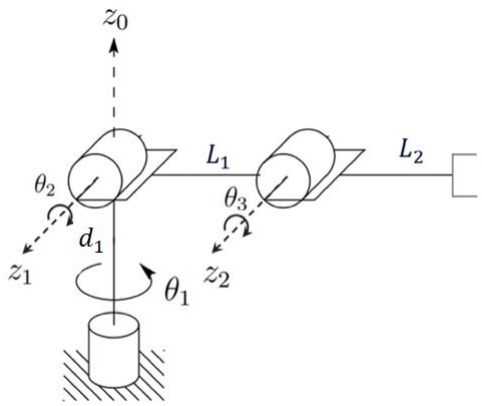


Fig. 1. Coordinate frames attached to RRR robot.

TABLE I
Link parameters for the RRR robot.

Joint	a_i	α_i	d_i	θ_i
1	0	$\frac{\pi}{2}$	d_1	$\theta_1{}^*$
2	L_1	0	0	$\theta_2{}^*$
3	L_2	0	0	$\theta_3{}^*$

The parameters in Table I are summarized as follows:

- Link Length a_i: Distance from the intersection of z_{i-1} and x_i axes to O_i.
- Link Twist α_i : Angle from z_{i-1} to z_i about x_i.
- Link Offset d_i : Distance from O_{i-1} to intersection of z_{i-1} and x_i.
- Joint Angle θ_i : The angle from x_{i-1} to x_i about z_{i-1}.

The transformation matrix T_i^{i-1} that connects every frame with the previous frame is giving by:

$$T_i^{i-1} = \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Based on that, the complete transformation between frames 3 and 0 is:

$$T_3^0 = T_1^0 T_2^1 T_3^2 = \begin{bmatrix} A & B & C & D \end{bmatrix}$$

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on October 29,2026 at 19:31:16 UTC from IEEE Xplore. Restrictions apply.

Where

$$A = \begin{bmatrix} \cos(\theta_2+\theta_3)\cos(\theta_1) \\ \cos(\theta_2+\theta_3)\sin(\theta_1) \\ \sin(\theta_2+\theta_3) \\ 0 \end{bmatrix}$$

$$B = \begin{bmatrix} -\sin(\theta_2+\theta_3)\cos(\theta_1) \\ -\sin(\theta_2+\theta_3)\sin(\theta_1) \\ \cos(\theta_2+\theta_3) \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} \sin(\theta_1) \\ -\cos(\theta_1) \\ 0 \\ 0 \end{bmatrix}$$

$$D = \begin{bmatrix} \cos(\theta_1)(L_2\cos(\theta_2+\theta_2)) + L_1\cos(\theta_2) \\ \sin(\theta_1)(L_2\cos(\theta_2+\theta_2)) + L_1\cos(\theta_2) \\ d_1 + L_2\sin(\theta_2+\theta_2) + L_1\sin(\theta_2) \\ 1 \end{bmatrix}$$

Figure 2 shows the CAD drawing of the RRR robot, providing a detailed representation of its mechanical structure, including the arrangement of joints, links, and end-effector. This drawing serves as a visual reference for participants, enabling them to analyze the robot's geometry and understand the relationships between its components. This Figure will be given to participants in order to solve for the D-H parameters table.

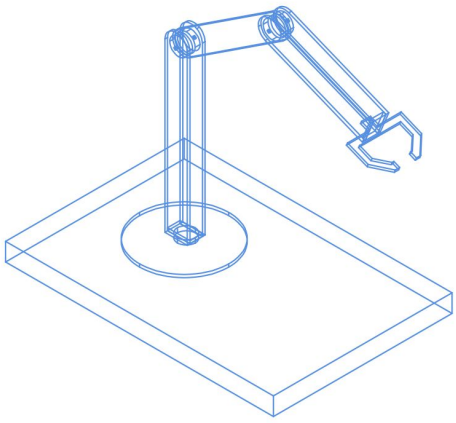


Fig. 2. CAD drawing of the RRR robot.

Figure 3 shows the model of the RRR robot with the attached frames. This highlights the D-H frames associated with each joint, allowing participants to grasp the concept of D-H parameters and their relevance to the robot's configuration.

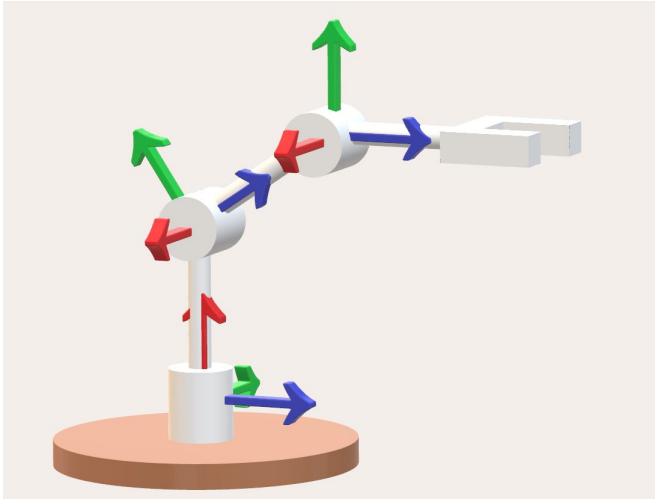


Fig. 3. RRR robot model with the attached frames.

To enhance the understanding of the robot's kinematics, a physical prototype of the RRR robot was developed for the survey, as shown in Figure 4. The prototype replicates the design and functionality of the CAD drawing, enabling participants to directly interact with the physical robot. This allows participants to explore the robot's movement, test their understanding of D-H parameters, and gain practical insights into its operation. The prototype was developed using household materials, resulting in a production cost of under $40.

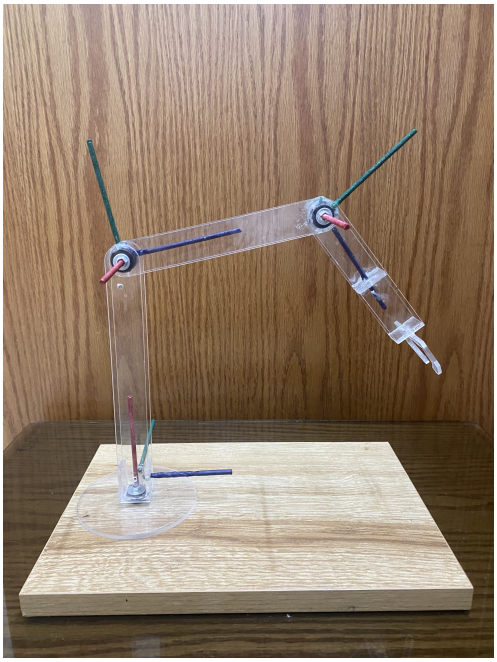


Fig. 4. The physical prototype of the RRR robot.

III. Findings

A. D-H Parameters Table

The primary objective of this study is to assess the level of understanding among students regarding D-H parameters

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on October 29,2026 at 19:31:16 UTC from IEEE Xplore. Restrictions apply.

in robotics using conventional teaching methods and the proposed prototype. We selected a sample of 36 students from the Mechatronics engineering department at the University of Jordan who attended the robotic systems course and 36 students who did not take the course yet.

In the study, students were asked to find the D-H parameters (Table I) of the RRR robot. Data were collected using a survey. This study consists of three parts: in the first part, the students found the D-H parameters by observing a CAD drawing of the RRR robot, and in the second part, they found the D-H parameters by observing a model of the RRR robot with frames attached to it. Finally, they found the D-H parameters by observing a prototype of the RRR robot.

Figure 5 shows the results for students who had not been enrolled in the course. Six students out of 36 were successful in figuring out the D-H parameters using the sketch, thirteen were successful using the model, and twenty-one were successful using the prototype.

It is important to observe that the majority (30 out of 36 students) preferred utilizing the prototype to find the D-H parameters, while only 6 students preferred using the sketch.

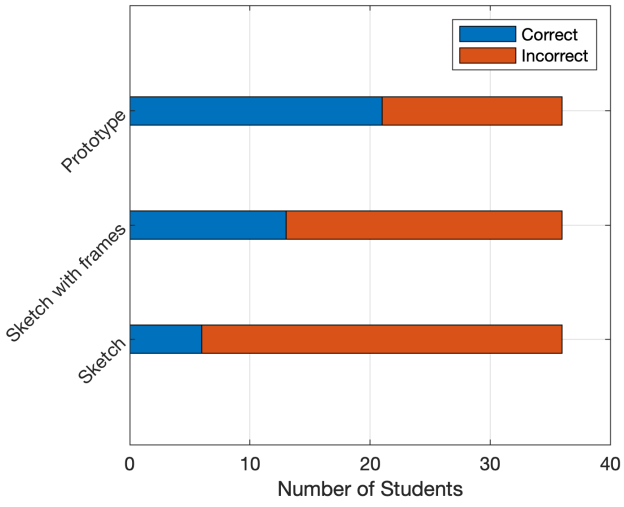


Fig. 5. Performance of students who did not enroll in the course.

Figure 6 shows the results for students who attended the course. Out of 36 students, ten were able to correctly determine the D-H parameters using the sketch, twenty-two were able to achieve so using the model, and 30 achieved it using the prototype.

The majority of the students (34 out of 36) indicated that the prototype enhanced the learning process and broadened their understanding of robot kinematics.

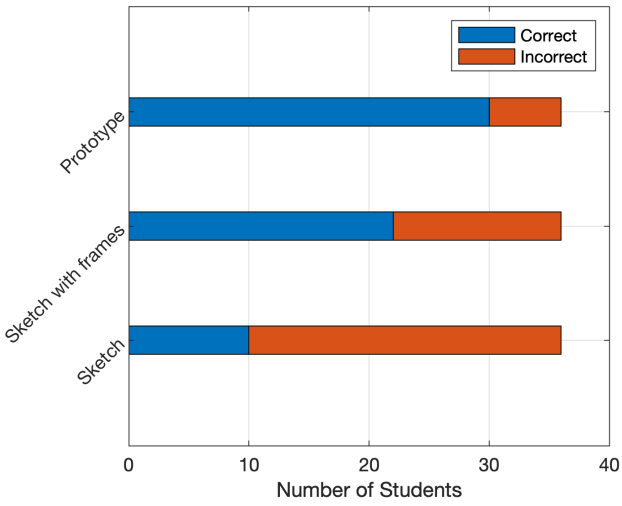


Fig. 6. Performance of students enrolled in the course.

Many students encountered challenges when determining θ_1, which represents the angle from x_0 to x_1 about z_0. A majority of the answers received for this parameter were either $0°$ or $90°$. This problem resulted from students' inability to imagine the joint's ability to move and alter the angle θ_1 when examining the sketch alone. As a result, students were attracted to the prototype as it provided a clearer understanding of the robot's joint movements and facilitated an easier identification of the D-H parameters. The prototype's interactive nature enabled students to observe the changes in θ_1 more effectively, leading to a higher preference for using the prototype to determine the D-H parameters.

Based on students' comments, the physical prototype allowed students to observe the four parameters of the D-H method in action while the robot is in motion. Unlike the static CAD drawing or the model with attached frames, the physical prototype enabled students to witness the parameters during the robot motion. By interacting with the moving robot, students can easily comprehend crucial aspects, such as the constant angle between two consecutive z-axes and the constant link offset, which may be challenging to discern solely from the CAD drawing or the model with attached frames. The physical prototype bridges the gap between theory and practice, facilitating a deeper understanding of the D-H parameters and their impact on the robot's kinematics.

B. Forward Kinematics

The second objective of this study is to assess the level of understanding among students regarding forward kinematics using conventional teaching methods and the proposed prototype.

The forward kinematics problem involves determining the position and orientation of the end-effector as a function of the joint angles. This can be achieved using the D-H parameters table, which provides a systematic approach to establishing the kinematic model of a robot. Additionally, in some cases,

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on October 29,2026 at 19:31:16 UTC from IEEE Xplore. Restrictions apply.

the forward kinematics can be found directly by visualizing the robot's motion.

In our study, we focused on finding the end-effector position (X, Y, Z) as a function of joint angles (θ_1, θ_2, and θ_3) for the RRR robot. We asked students to find the forward kinematics equations using the model in Figure 3 and the prototype in Figure 4. We did not include the CAD drawing as it is hard to visualize the forward kinematics model using only the CAD drawing. The results of this analysis are summarized in Table II. The percentages in Table II represent the students' correct answers.

TABLE II
FORWARD KINEMATICS ANALYSIS.

		Correct answers given by students who took the course	Correct answers given by students who did not take the course
X	Model	25%	5%
	Prototype	60%	12%
Y	Model	27%	4%
	Prototype	60%	14%
Z	Model	35%	10%
	Prototype	75%	18%

When visualizing the forward kinematics equations, the majority of students indicated a preference for the physical prototype, according to the results presented in Table II. The interactive nature of the prototype enabled students to observe the robot's motion and understand its kinematics, making it an effective tool for visualizing forward kinematics equations.

In the context of industrial robotics courses, it is often essential for students to gain practical experience with various well-known industrial robots. However, the cost of acquiring and maintaining a wide range of industrial robots can be costly for some universities. To address this challenge, the development of a low-cost manipulator model with frames can prove invaluable in enhancing the learning process and providing students with hands-on experience.

IV. CONCLUSIONS

Our study successfully constructed a physical prototype using household materials at a cost of less than $40. This low-cost approach makes it possible for universities with limited resources to incorporate manipulator models into their industrial robotics courses. The manipulator model, equipped with frames representing the D-H parameters, offers students an opportunity to analyze and understand the kinematics of different industrial robots. It also enables students to study and compare the kinematics of various robots, building a solid foundation of knowledge and practical skills in the field of industrial robotics. As a result, incorporating low-cost manipulator models serves as a valuable addition to industrial robotics courses, enhancing the learning process and broadening students' understanding of robot kinematics.

REFERENCES

[1] D. Scaradozzi, L. Sorbi, A. Pedale, M. Valzano, and C. Vergine, "Teaching robotics at the primary school: an innovative approach," *Procedia-Social and Behavioral Sciences*, vol. 174, pp. 3838–3846, 2015.

[2] L. Slangen, H. van Keulen, and K. Gravemeijer, "Preparing teachers to teach robotics in primary schools," in *Professional development for primary teachers in science and technology*. Brill, 2011, pp. 181–198.

[3] J. W. van der Molen, J. de Lange, and J. Kok, "Theoretische uitgangspunten bij de professionalisering van leraren basisonderwijs op het gebied van wetenschap en techniek," in *Onderzoek naar wetenschap en techniek in het Nederlandse basisonderwijs*. Platform Bèta Techniek, 2009, pp. 29–39.

[4] S. González-García, J. Rodríguez-Arce, G. Loreto-Gómez, and V. M. Montaño-Serrano, "Teaching forward kinematics in a robotics course using simulations: Transfer to a real-world context using lego mindstorms™," *International Journal on Interactive Design and Manufacturing (IJIDeM)*, vol. 14, pp. 773–787, 2020.

[5] S. E. Jung and E.-s. Won, "Systematic review of research trends in robotics education for young children," *Sustainability*, vol. 10, no. 4, p. 905, 2018.

[6] M. Akçayır and G. Akçayır, "Advantages and challenges associated with augmented reality for education: A systematic review of the literature," *Educational research review*, vol. 20, pp. 1–11, 2017.

[7] K. Highfield, "Robotic toys as a catalyst for mathematical problem solving," *Australian primary mathematics classroom*, vol. 15, no. 2, pp. 22–27, 2010.

[8] B. Karlik and S. Aydin, "An improved approach to the solution of inverse kinematics problems for robot manipulators," *Engineering applications of artificial intelligence*, vol. 13, no. 2, pp. 159–164, 2000.

[9] R. S. Othayoth, R. G. Chittawadigi, R. P. Joshi, and S. K. Saha, "Robot kinematics made easy using roboanalyzer software," *Computer Applications in Engineering Education*, vol. 25, no. 5, pp. 669–680, 2017.

[10] R. Arnay, J. Hernández-Aceituno, E. González, and L. Acosta, "Teaching kinematics with interactive schematics and 3d models," *Computer Applications in Engineering Education*, vol. 25, no. 3, pp. 420–429, 2017.

[11] S. Kvesko, N. Kvesko, A. Korniyenko, and N. N. Kabanova, "Educational robotics as an innovative teaching practice using technology: minimization of risks," in *IOP Conference Series: Materials Science and Engineering*, vol. 363, no. 1. IOP Publishing, 2018, p. 012004.

[12] G. Loreto-Gómez, J. Rodríguez-Arce, S. González-García, and V. M. Montaño-Serrano, "Analysing the effect of the use of 3d simulations on the performance of engineering students in a robotics course: findings from a pilot study," *The International Journal of Electrical Engineering & Education*, vol. 56, no. 2, pp. 163–178, 2019.

[13] A. Gil, O. Reinoso, J. M. Marin, L. Paya, and J. Ruiz, "Development and deployment of a new robotics toolbox for education," *Computer Applications in Engineering Education*, vol. 23, no. 3, pp. 443–454, 2015.

[14] J. Bahuguna, R. G. Chittawadigi, and S. K. Saha, "Teaching and learning of robot kinematics using roboanalyzer software," in *Proceedings of Conference on Advances In Robotics*, 2013, pp. 1–6.

[15] J. A. N. Cocota, H. S. Fujita, and I. J. da Silva, "A low-cost robot manipulator for education," in *2012 Technologies Applied to Electronics Teaching (TAEE)*. IEEE, 2012, pp. 164–169.

[16] F. Arenas-Rosales, F. Martell-Chavez, I. Y. Sanchez-Chavez, and C. A. Paredes-Orta, "Virtual ur5 robot for online learning of inverse kinematics and independent joint control validated with fsm position control," *Robotics*, vol. 12, no. 1, p. 23, 2023.

[17] A. L. Talli, V. K. V. Meti, and A. C. Giriyapur, "An effective way of teaching fundamentals of robotics to undergraduate students using roboanalyzer," *Journal of Engineering Education Transformations*, vol. 35, no. 3, 2022.

[18] J. Bertram, J. Moskaliuk, and U. Cress, "Virtual training: Making reality work?" *Computers in Human Behavior*, vol. 43, pp. 284–292, 2015.

[19] M. E. Dankbaar, J. Alsma, E. E. Jansen, J. J. van Merrienboer, J. L. van Saase, and S. C. Schuit, "An experimental study on the effects of a simulation game on students' clinical cognitive skills and motivation," *Advances in Health Sciences Education*, vol. 21, pp. 505–521, 2016.

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on October 29,2026 at 19:31:16 UTC from IEEE Xplore. Restrictions apply.